

Session 1 — Where things are: the celestial sphere

Practical Astronomy Intensive

Name _____

Date _____

9 questions 15 minutes

Nine questions at the end of the first indoor session. They arrived able to define the celestial sphere and unable to point at the meridian; this checks whether the definitions have turned into something they could act on outside tonight. Angular measure is asked twice — everything later leans on it and nobody arrives with it.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

- 1 You hold a clenched fist out at arm's length against the sky. Roughly what angle does it span, and how does the full Moon compare?
 - a About 45° — and the full Moon is roughly a coin's width at arm's length
 - b About 1° — and the full Moon is a few times larger than that
 - c About 10° — and the full Moon is far smaller, about half a degree, covered by a little fingernail at arm's length
 - d About 10° — and the full Moon is about the same, roughly a fist's width
- 2 Mizar and Alcor sit about 12 arcminutes apart in the handle of the Big Dipper. A friend says 'so they must be practically touching out there.' What is wrong with that?
 - a It is wrong only because 12 arcminutes is actually a large angle, so they are far apart in space
 - b Angular separation is a difference in direction only — it says nothing about how far apart the two stars actually are in space
 - c The Sun and Moon both span half a degree, so angles are the same for every object and mean nothing
 - d Nothing is wrong — 12 arcminutes is a tiny angle, so they must be close together in space
- 3 A long-exposure photograph shows the stars drawn out into concentric arcs around a single point. What has the photograph actually recorded?
 - a The camera, bolted to the Earth, turning eastward through the exposure — the arcs are the Earth's rotation drawn out
 - b The stars orbiting the pole star, which sits at the centre of their motion
 - c The stars themselves sailing westward across the sky during the night
 - d Every star in the frame rising and setting, since everything visible rises and sets

- 4 What is the celestial sphere, and where do its poles and equator come from?
- a A bookkeeping model of infinite radius that turns directions into geometry — its poles and equator are Earth's own axis and equator projected outward
 - b A real shell at a great distance with the stars fixed on it, all at the same distance from us
 - c Another name for the sky dome — the visible half of the sky above your horizon
 - d A model sphere, but with its poles and equator being genuine features of the sky, independent of Earth
- 5 A student writes in her log: 'Jupiter — altitude 34° , azimuth 118° .' What must she add for that entry to be usable by anyone, including her?
- a Nothing — altitude and azimuth are a fixed address for an object, like RA and Dec
 - b Only the date — azimuth is fixed for an object and only the altitude drifts
 - c The direction she was facing when she measured, since azimuth runs from there
 - d The date, time and her location — alt-az changes minute by minute and is meaningless without all three
- 6 You are standing in Bhopal (latitude 23° N) and want to show a group Polaris. Which statement is true, and is the most useful thing to tell them?
- a It is a middling star, about 48th brightest — find it with the Pointers of Ursa Major, only about 23° above the northern horizon
 - b It is the brightest star in the sky — just look north and pick out the brightest thing there
 - c It is the star nearest to Earth, which is why it barely moves in the sky
 - d It sits almost directly overhead, so look up near the zenith
- 7 A student in Bhopal (23° N) finds that Ursa Major dips below his northern horizon in the evening. A friend in Srinagar (34° N) insists it never sets for her. Who is right?
- a Both — which stars are circumpolar depends only on your latitude, and Srinagar's higher pole keeps more of Ursa Major above the horizon
 - b The Bhopal student — every star rises and sets, so nothing can truly be circumpolar
 - c The Srinagar student — Ursa Major is circumpolar, and that is a fixed property of the constellation for everybody
 - d Neither — circumpolarity depends on the season, and they simply observed at different times of year

8 Saturn is up all night. Why is it worth waiting for it to transit your meridian rather than observing it at 9 pm when it is low in the east?

- a At transit it is at its highest, so you look through the least air — low in the east the same planet is a shimmering mush
- b Because the meridian is a line printed on the star chart, and objects are only in the catalogue's position when they cross it
- c There is no real difference — an object is equally worth observing at any time it happens to be above the horizon
- d Because at transit the planet crosses the celestial equator, which is where the atmosphere is thinnest

9 An astronomer in London writes to a student in Chennai asking her to describe Canopus, the second-brightest star in the sky, since he has never seen it. Why has he never seen it?

- a Canopus is only visible from the southern hemisphere, and Chennai is in the southern hemisphere
- b London is too far north for stars to rise and set at all — everything there just circles the pole
- c He must simply have been unlucky with weather or timing — everyone on Earth sees the same stars eventually
- d Canopus is far enough south that it never rises for London's latitude, while from Chennai it is a routine sight